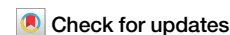


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Machine learning predicts sepsis deterioration trajectories



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Sepsis has heterogeneous clinical trajectories, but conventional severity scores offer only static risk estimates. Timely, dynamic prediction could enable personalized intervention. In this multicenter retrospective study of 47,936 ICU patients meeting Sepsis-3 criteria from one institutional and two public datasets (MIMIC-III, eICU; sensitivity in MIMIC-IV), group-based trajectory modeling identified latent recovery patterns. An ensemble machine-learning model incorporating dynamic physiological variability was trained, temporally validated, and externally tested; clinical impact was assessed following implementation. Three trajectories emerged: rapid recovery (41.5%), slow recovery (36.4%), and clinical deterioration (22.1%). In the final binary classification task, AUROC was 0.92 (development), 0.89 (internal), 0.84 (MIMIC-III) and 0.77 (eICU); median warning time before deterioration was 17.6 h (Overall pooled across all cohorts). Reduced heart rate variability (SD < 10 bpm) predicted mortality (adjusted HR 2.17). Implementation reduced ICU stay by 1.8 days, mechanical ventilation by 2.3 days, and 28-day mortality by 5.7%. This externally validated trajectory-based model offers accurate, early risk stratification for sepsis, supporting proactive, individualized critical care.

Sepsis, a life-threatening organ dysfunction caused by a dysregulated host response to infection, affects approximately 49 million patients globally each year, resulting in 11 million deaths and accounting for nearly 20% of all global deaths¹. Despite advances in diagnosis and treatment, sepsis remains a leading cause of mortality in intensive care units (ICUs), with survivors often facing long-term functional impairments and reduced quality of life^{2,3}.

The clinical progression of sepsis exhibits significant heterogeneity, with patients potentially experiencing rapid recovery, gradual improvement, or continued deterioration⁴. Traditional severity scoring systems, such as the Sequential Organ Failure Assessment (SOFA) and Acute Physiology and Chronic Health Evaluation II (APACHEII), provide only static assessments and struggle to capture the dynamic evolution of the disease^{5,6}. Furthermore, these scoring systems typically focus on mortality prediction rather than the dynamic changes in patient recovery trajectories^{7,8}.

In recent years, machine-learning techniques have made significant advances in clinical prediction^{9,10}. Compared with traditional statistical methods, machine learning can handle high-dimensional data, identify complex nonlinear relationships, and integrate time-series information¹¹. However, existing studies have largely focused on binary outcome

prediction (e.g., mortality) and less on dynamic prediction of patient recovery trajectories^{12,13}. Additionally, many prediction models lack external validation, particularly in real-world clinical settings^{14,15}.

This study aimed to develop and validate a machine-learning model for predicting recovery trajectories and identifying deterioration risk in patients with sepsis. We employed a multi-level validation strategy, combining institutional data with large public databases (MIMIC-III and eICU) to evaluate model performance across different healthcare settings. We paid special attention to temporal variability analysis of physiological parameters, exploring their potential as a new perspective for prognostic prediction. Finally, we assessed the impact of trajectory prediction-based individualized intervention strategies on clinical outcomes.

Results

Cohort development and patient characteristics

The study utilized a multicenter sepsis database comprising 54,758 patients, with 6822 excluded due to missing data (>20%, $n = 2541$), ICU stays <24 h ($n = 1873$), transfers from external hospitals ($n = 1248$), or pre-existing limitations of care ($n = 1160$). Excluded patients ($n = 2541$) had similar baseline

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characteristics to included cases ($SMD < 0.2$ for all variables), minimizing selection bias. The final analysis cohort included 47,936 patients, divided into development ($n = 2843$), internal validation ($n = 1213$), and external validation cohorts (MIMIC-III: $n = 25,633$; eICU: $n = 18,247$) (Fig. 1).

Table 1 shows the baseline characteristics of patients in the development and internal validation cohorts. The median age was 63.8 years (IQR, 51.9–75.4), and 2352 (58.0%) were male. The most common sources of infection were pulmonary (42.1%), followed by abdominal (23.8%) and urinary tract (16.5%). The median baseline SOFA score was 7 (IQR, 5–10). There were no significant differences between the development and validation cohorts in terms of age, sex distribution, comorbidity burden, or baseline severity scores. However, the validation cohort had a slightly higher proportion of patients with pulmonary infections (44.3% vs 41.2%, $P = 0.04$) and a lower proportion with abdominal infections (21.9% vs 24.6%, $P = 0.03$).

External validation cohort characteristics

Supplementary Table 1 presents the baseline characteristics of the external validation cohorts compared with the development cohort. MIMIC-III cohort patients were slightly older (median age 65.7 years vs. 63.8 years, $SMD = 0.18$) and had a higher proportion of pulmonary infections (47.3% vs. 42.1%, $SMD = 0.10$). The eICU cohort had a higher proportion of urinary tract infections (19.8% vs. 16.5%, $SMD = 0.09$) and a lower baseline SOFA score (median 6 vs. 7, $SMD = 0.12$). All three cohorts showed similar sex distribution and comorbidity burden ($SMD < 0.1$).

Trajectory identification and clinical patterns

Group-based multi-trajectory modeling identified three distinct sepsis trajectories (Fig. 2a–d; Supplementary Fig. 1a):

1. Rapid Recovery (41.5%, $n = 19,883$): Characterized by a third-order polynomial decline in SOFA scores ($\beta_3 = -0.0023$, $P < 0.001$), with median SOFA reduction of 3 points (IQR: 2–4) within 48 h, early hemodynamic stabilization, and lactate normalization.

2. Slow Recovery (36.4%, $n = 17,449$): Exhibited a second-order polynomial SOFA decline ($\beta_2 = -0.0018$, $P < 0.001$), median SOFA reduction of 1 point (IQR: 0–2), and gradual organ function improvement.

3. Clinical Deterioration (22.1%, $n = 10,604$): Demonstrated ascending SOFA trends ($\beta_2 = +0.0031$, $P < 0.001$), median SOFA increases of 2 points (IQR: 1–4), and escalating organ support needs.

All mean trajectory plots in Supplementary Fig. 2a include 95% confidence intervals, and individual patient trajectories are visualized in Supplementary Fig. 2b, demonstrating robust group separation with less than 5% trajectory crossover.

Patients in the rapid recovery group experienced a median SOFA score decrease of 3 points (IQR, 2–4) within 48 h, accompanied by rapid hemodynamic stabilization and lactate normalization. The slow recovery group showed a median SOFA score decrease of 1 point (IQR, 0–2) within 48 h with gradual improvement in organ function. The clinical deterioration group exhibited a median SOFA score increase of 2 points (IQR, 1–4), typically requiring escalation of organ support (Fig. 2b).

Incorporating PROC TRAJ posterior probabilities as predictive variables into machine-learning models significantly enhanced model performance (Supplementary Fig. 1b). In the development cohort, AUROC improved from 0.82 (95% CI 0.80–0.84) to 0.84 (95% CI 0.82–0.86), and in the validation cohort from 0.81 (95% CI 0.78–0.84) to 0.83 (95% CI 0.80–0.86). Mortality risk differences between trajectory groups are shown in Supplementary Fig. 1c.

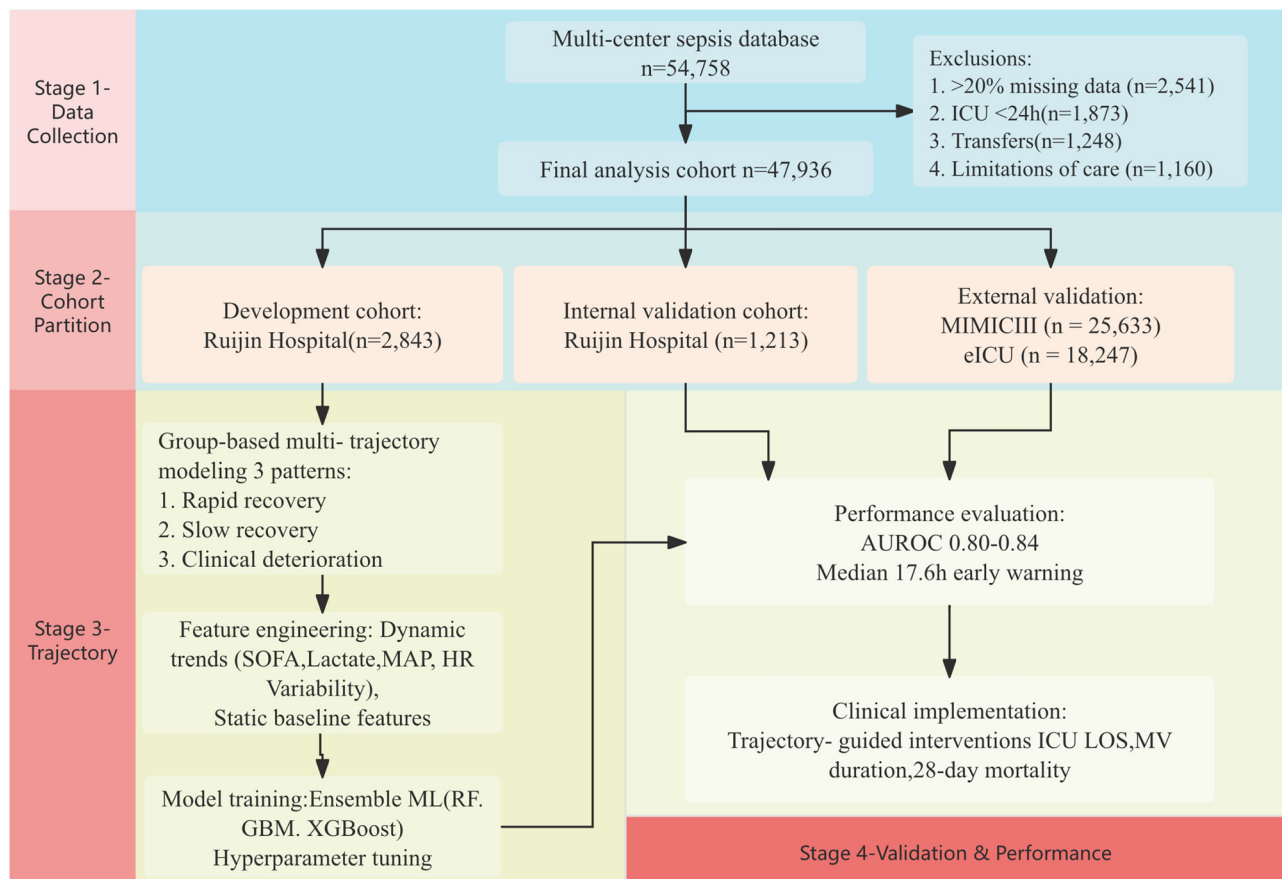


Fig. 1 | Study design and methodology. This figure illustrates the three-phase study approach (trajectory identification, model development, and implementation) and patient cohort development. It shows how the initial multi-center sepsis database ($n = 54,758$) was refined to a final analysis cohort ($n = 47,936$) after exclusions, and

further divided into development, validation, and implementation cohorts. ICU Intensive Care Unit, EHR Electronic Health Record, MIMIC-III Medical Information Mart for Intensive Care III, eICU Electronic Intensive Care Unit.

Table 1 | Baseline characteristics of patients in development and validation cohorts

Characteristic	Development Cohort (n = 2843)	Internal Validation Cohort (n = 1213)	P Value
Demographics			
Age, median (IQR), y	63.8 (51.3–75.1)	64.2 (52.6–75.8)	0.21
Male, No. (%)	1648 (58.0)	704 (58.0)	0.98
BMI, median (IQR)	24.3 (21.8–27.2)	24.1 (21.6–27.0)	0.34
Comorbidities, No. (%)			
Hypertension	1251 (44.0)	542 (44.7)	0.69
Diabetes	768 (27.0)	336 (27.7)	0.66
Coronary artery disease	597 (21.0)	261 (21.5)	0.72
Congestive heart failure	370 (13.0)	164 (13.5)	0.67
COPD	455 (16.0)	200 (16.5)	0.70
Chronic kidney disease	427 (15.0)	188 (15.5)	0.69
Cancer	512 (18.0)	224 (18.5)	0.72
Charlson Comorbidity Index, median (IQR)	3 (2–5)	3 (2–5)	0.56
Infection Source, No. (%)			
Pulmonary	1171 (41.2)	537 (44.3)	0.04
Abdominal	699 (24.6)	266 (21.9)	0.03
Urinary tract	469 (16.5)	197 (16.2)	0.82
Skin/soft tissue	228 (8.0)	95 (7.8)	0.84
Bloodstream	171 (6.0)	75 (6.2)	0.82
Other/unknown	105 (3.7)	43 (3.5)	0.78
Baseline Severity			
SOFA score, median (IQR)	7 (5–10)	7 (5–10)	0.61
APACHE II score, median (IQR)	18 (14–24)	19 (14–25)	0.29
Lactate, median (IQR), mmol/L	2.4 (1.6–3.8)	2.5 (1.7–3.9)	0.32
Mechanical ventilation, No. (%)	1508 (53.0)	655 (54.0)	0.58
Vasopressor use, No. (%)	1365 (48.0)	594 (49.0)	0.58
Laboratory Values, median (IQR)			
White blood cell count, 10 ³ /μL	12.8 (8.4–18.2)	13.1 (8.6–18.5)	0.28
Platelet count, 10 ³ /μL	186 (118–265)	182 (115–260)	0.36
Creatinine, mg/dL	1.3 (0.9–2.1)	1.3 (0.9–2.2)	0.68
Total bilirubin, mg/dL	1.1 (0.6–2.3)	1.2 (0.6–2.4)	0.42
C-reactive protein, mg/L	145 (86–224)	151 (90–230)	0.19
Outcomes			
ICU length of stay, median (IQR), d	6.2 (3.1–12.4)	6.5 (3.3–12.8)	0.24
Hospital length of stay, median (IQR), d	15.4 (9.2–26.7)	16.1 (9.5–27.3)	0.18
Mechanical ventilation duration, median (IQR), d	4.8 (2.3–9.6)	5.0 (2.5–9.8)	0.31
ICU mortality, No. (%)	512 (18.0)	225 (18.5)	0.70
28-day mortality, No. (%)	626 (22.0)	273 (22.5)	0.73

APACHE Acute Physiology and Chronic Health Evaluation, BMI body mass index, COPD chronic obstructive pulmonary disease, ICU intensive care unit, IQR interquartile range, SOFA Sequential Organ Failure Assessment.

Principal component analysis (PCA) revealed clear separation among trajectory groups (Fig. 2a). Temporal trends in clinical parameters further distinguished trajectories: the deterioration group displayed persistently elevated lactate (median peak: 3.4 mmol/L vs. 1.8 mmol/L in rapid recovery), reduced heart rate variability (HR SD: 8.4 bpm vs. 14.2 bpm), and

progressive respiratory instability (Fig. 2c; Supplementary Fig. 3, where Group 1 = Rapid Recovery, Group 2 = Slow Recovery, Group 3 = Clinical Deterioration). SOFA slope analysis demonstrated that rapid decline (>2 points/24 h) correlated with reduced mortality (adjusted HR: 0.67, 95% CI: 0.58–0.78), while increases (>1 point/24 h) heightened risk (HR: 1.92, 95% CI: 1.65–2.24) (Supplementary Fig. 1d).

Table 2 presents the baseline characteristics and outcomes across the three trajectory groups. Patients in the clinical deterioration group were older (median age 68.4 vs 61.3 and 63.5 years, $P < 0.001$), had higher comorbidity burdens (median Charlson Comorbidity Index 4 vs 2 and 3, $P < 0.001$), and higher baseline SOFA scores (median 9 vs 6 and 7, $P < 0.001$) compared with the rapid and slow recovery groups, respectively.

Trajectory slope analysis revealed that SOFA score decline rates >2 points per 24 h associated with significantly improved survival (adjusted HR 0.67, 95% CI 0.58–0.78), while increase rates >1 point per 24 h correlated with significantly increased mortality risk (adjusted HR 1.92, 95% CI 1.65–2.24). Notably, we found that trajectory slope changes (acceleration) served as important prognostic indicators, suggesting disease progression dynamics hold greater predictive value than point assessments.

Model performance and early warning capability

As shown in Fig. 2, several dynamic physiological parameters—including lactate clearance rate, heart rate variability, SOFA score trend, mean arterial pressure variability, and respiratory rate variability—displayed marked differences between the identified sepsis trajectories. In the final prediction model, these five variables were included among the broader feature set and ranked within the top contributors in SHAP importance analysis (Fig. 3c). Their prominence in model interpretability reflects the fact that trajectory-based differences observed in Fig. 2 were not only visually apparent but also statistically and prognostically significant, reinforcing the consistency between trajectory discovery and model-based prediction. The ensemble learning model demonstrated strong discrimination in the final binary classification task (clinical deterioration vs. non-deterioration), with AUROC values of 0.92 (95% CI: 0.91–0.93) in the development cohort, 0.89 (95% CI: 0.87–0.91) in internal validation, 0.84 (95% CI: 0.83–0.85) in the MIMIC-III cohort, and 0.77 (95% CI: 0.75–0.78) in the eICU cohort (Fig. 3; Table 4). Corresponding AUPRC values were 0.87, 0.84, 0.80, and 0.73, respectively. Sensitivity and specificity at the optimal Youden Index threshold ranged from 0.82–0.84 to 0.79–0.86, with Brier scores between 0.098 and 0.118, indicating strong discrimination and good calibration across all datasets.

Figure 3a presents ROC curves for all study cohorts, with AUC values matching those in Table 4; Fig. 3b presents Precision–Recall curves for the same datasets, showing corresponding AUPRC values. This separation ensures clarity in metric interpretation and aligns the curves with the quantitative results. Table 3 summarizes the performance of the trajectory classification task at different time points after ICU admission. In the development cohort, AUROC increased from 0.79 at 6 h to 0.84 at 24 h, with similar trends in internal and external validations. AUPRC values also improved over time, and calibration remained good (Brier scores 0.10–0.16). These results highlight the model's capability to dynamically classify recovery trajectories during the ICU stay. By presenting both Tables 3 and 4, we clarify that the trajectory classification task evaluates the model's longitudinal predictive ability, whereas the binary classification task assesses single-window discrimination performance. This dual presentation eliminates potential inconsistencies between table values and Fig. 3 and underscores complementary strengths of the prediction framework.

These calibration metrics are presented separately for binary classification (Supplementary Fig. 4a) and trajectory classification (Supplementary Fig. 5) to avoid cross-task comparison. Further validating that the model's risk estimates align with real-world outcomes—whether for the binary deterioration task or trajectory classification task. Decision curve analysis indicated substantial net benefit across a range of probability thresholds, supporting clinical utility (Supplementary Fig. 4d).

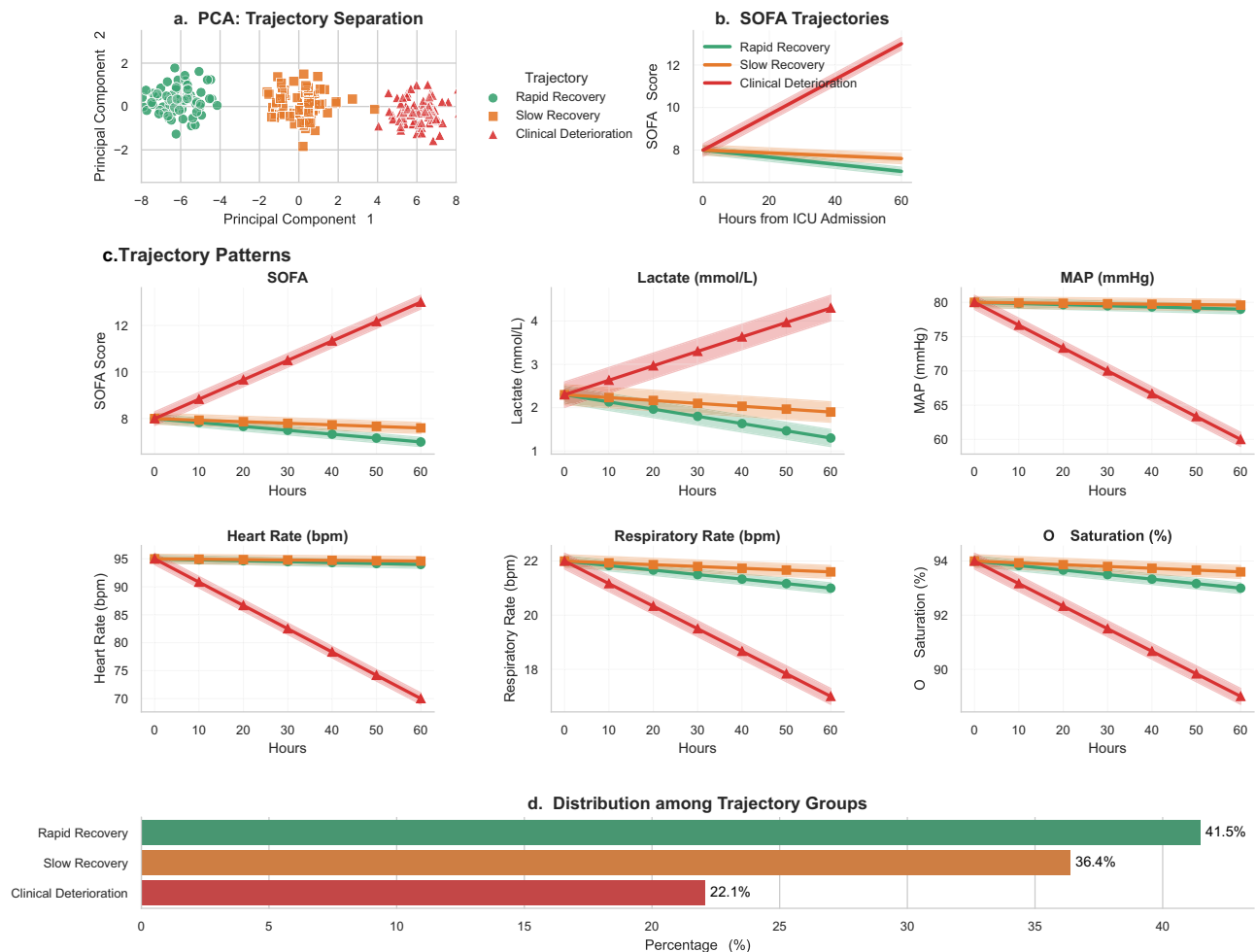


Fig. 2 | Sepsis trajectory patterns and clinical parameters. **a** Principal component analysis (PCA) shows the separation of patients according to their sepsis trajectories: Rapid Recovery, Slow Recovery, and Clinical Deterioration. **b** Representative longitudinal SOFA score patterns for each trajectory group over the first 60 h of ICU admission. **c** Key clinical parameters over time for each trajectory group: SOFA score, lactate, mean arterial pressure (MAP), heart rate, respiratory rate, and oxygen saturation; shaded areas represent $\pm 95\%$ CI around each trajectory curve. Several

parameters with prominent inter-trajectory differences (lactate clearance, HR variability, SOFA trend, MAP variability, RR variability) were further evaluated in the final prediction model. **d** Distribution of the study population across the three identified trajectory groups. SOFA Sequential Organ Failure Assessment, MAP Mean Arterial Pressure, ICU Intensive Care Unit, bpm Beats Per Minute (heart rate) or Breaths Per Minute (respiratory rate), mmHg Millimeters of Mercury- mmol/L Millimoles per Liter.

The model provided a median warning time of 17.6 h (Overall pooled across all cohorts IQR 11.7–23.5; Development: 18.6, Internal: 18.6, MIMIC-III: 17.8, eICU: 16.5) before clinical deterioration, with sensitivity and specificity of 0.83 (95% CI: 0.81–0.85) and 0.88 (95% CI: 0.86–0.90), respectively, within the critical 12-hour prediction window. Subgroup analyses confirmed consistent performance across infection sources (pulmonary AUROC 0.85 vs. urinary tract 0.83), age groups (<65 years AUROC 0.85 vs. ≥ 65 years AUROC 0.83), and baseline SOFA scores (≥ 7 AUROC 0.85 vs. <7 AUROC 0.82) (Supplementary Table 2).

Feature Importance and Physiological Variability

SHAP analysis identified dynamic parameters as top predictors: lactate clearance rate (mean SHAP value: 0.32), heart rate variability (0.28), and SOFA trend (0.25) (Fig. 3c; Supplementary Fig. 4c). Across all cohorts, trajectory group, SOFA slope, and comorbidity index consistently ranked among the highest contributors, reflecting the strong influence of temporal disease progression on deterioration risk. Reduced heart rate variability (HR SD < 10 bpm) was associated with a 2.17-fold mortality risk (adjusted HR: 2.17, 95% CI: 1.68–2.79) (Supplementary Fig. 6c). Temporal variability analysis revealed a progressive loss of HR complexity 12–24 h before overt decline (Supplementary Fig. 6b), aligning with the “critical slowing down” hypothesis. This association remained consistent across validation cohorts

(development: HR = 2.17; MIMIC-III: HR = 2.09; eICU: HR = 1.94) (Supplementary Fig. 6d). Multivariable Cox analysis confirmed trajectory group, SOFA slope, and physiological variability as independent mortality predictors (Supplementary Table 3). SHAP interaction analysis further identified SOFA slope $\times$ lactate (Δ AUC + 0.07) and HR variability $\times$ vasopressor use (Δ AUC + 0.05) as the most synergistic feature pairs (Supplementary Fig. 7, Supplementary Fig. 8).

Trajectory classification performance

In the multi-time-point trajectory classification task (Table 3), the model showed steadily improving discrimination as more clinical data became available after ICU admission. In the development cohort, AUROC rose from 0.79 (95% CI: 0.77–0.81) at 6 h to 0.84 (95% CI: 0.82–0.86) at 24 h, with similar improvements in internal validation (0.78 to 0.83), MIMIC-III (0.77 to 0.82), and eICU cohorts (0.75 to 0.78). AUPRC values increased in parallel, and calibration was consistently good (trajectory classification task; Brier scores declining from 0.14 to 0.10 in the development cohort; Supplementary Fig. 5).

For patients predicted to follow the clinical deterioration trajectory, the second-stage timing model provided a median warning time of 17.6 h (IQR: 11.7–23.5) before onset, with 12-h window sensitivity and specificity of 0.83 and 0.88 in internal validation, and comparable values across external cohorts.

Table 2 | Baseline characteristics and clinical outcomes by trajectory groups in the combined cohort (n = 47,936)

Characteristic	Rapid Recovery (n = 19,883)	Slow Recovery (n = 17,449)	Clinical Deterioration (n = 10,604)	P Value
Demographics				
Age, median (IQR), y	61.3 (49.2–72.4)	63.5 (51.1–74.8)	68.4 (56.9–79.2)	<0.001
Male, No. (%)	11,532 (58.0)	10,121 (58.0)	6,151 (58.0)	0.99
BMI, median (IQR)	25.8 (22.6–29.7)	25.9 (22.5–30.1)	26.1 (22.4–30.8)	0.15
Comorbidities				
Charlson Comorbidity Index, median (IQR)	2 (1–4)	3 (2–5)	4 (2–6)	<0.001
Hypertension, No. (%)	8548 (43.0)	8027 (46.0)	5514 (52.0)	<0.001
Diabetes, No. (%)	5171 (26.0)	5060 (29.0)	3712 (35.0)	<0.001
Coronary artery disease, No. (%)	3977 (20.0)	4188 (24.0)	3287 (31.0)	<0.001
Congestive heart failure, No. (%)	2386 (12.0)	3141 (18.0)	2756 (26.0)	<0.001
COPD, No. (%)	2983 (15.0)	3141 (18.0)	2121 (20.0)	<0.001
Chronic kidney disease, No. (%)	2783 (14.0)	3141 (18.0)	2439 (23.0)	<0.001
Infection Source, No. (%)				
Pulmonary	8749 (44.0)	7677 (44.0)	4877 (46.0)	0.001
Abdominal	4572 (23.0)	4013 (23.0)	2227 (21.0)	<0.001
Urinary tract	3580 (18.0)	2967 (17.0)	1803 (17.0)	0.01
Skin/soft tissue	1590 (8.0)	1396 (8.0)	742 (7.0)	0.01
Bloodstream	1193 (6.0)	1047 (6.0)	636 (6.0)	0.99
Other/unknown	199 (1.0)	349 (2.0)	318 (3.0)	<0.001
Baseline Severity				
SOFA score, median (IQR)	6 (4–8)	7 (5–9)	9 (7–12)	<0.001
APACHE II score, median (IQR)	16 (12–21)	18 (14–23)	22 (17–28)	<0.001
Lactate, median (IQR), mmol/L	1.8 (1.3–2.7)	2.3 (1.6–3.5)	3.4 (2.1–5.8)	<0.001
Mechanical ventilation, No. (%)	8749 (44.0)	9597 (55.0)	7423 (70.0)	<0.001
Vasopressor use, No. (%)	7159 (36.0)	8725 (50.0)	7211 (68.0)	<0.001
Laboratory Values, median (IQR)				
White blood cell count, 10 ³ /μL	12.1 (8.2–16.8)	12.7 (8.4–18.1)	13.6 (8.6–19.7)	<0.001
Platelet count, 10 ³ /μL	208 (152–278)	187 (124–264)	162 (96–248)	<0.001
Creatinine, mg/dL	1.1 (0.8–1.7)	1.3 (0.9–2.1)	1.6 (1.0–2.7)	<0.001
Total bilirubin, mg/dL	0.8 (0.5–1.4)	1.0 (0.6–2.0)	1.5 (0.7–3.2)	<0.001
C-reactive protein, mg/L	132 (78–206)	146 (87–225)	168 (96–248)	<0.001
Physiological Variability Metrics				
Heart rate SD, median (IQR), bpm	14.2 (10.8–18.7)	11.6 (8.4–15.3)	8.4 (5.7–11.8)	<0.001
Mean arterial pressure SD, median (IQR), mmHg	12.8 (9.6–16.5)	10.3 (7.5–13.8)	8.1 (5.2–11.3)	<0.001
Respiratory rate SD, median (IQR), breaths/min	5.2 (3.8–7.1)	4.8 (3.5–6.7)	4.2 (2.9–6.5)	<0.001
Trajectory Characteristics				
SOFA change at 48 h, median (IQR), points	−3 (−4 to −2)	−1 (−2 to 0)	+2 (+1 to +4)	<0.001
Lactate clearance at 24 h, median (IQR), %	42.3 (28.5–58.7)	24.6 (12.3–38.4)	−15.8 (−42.6 to +8.3)	<0.001
Clinical Outcomes				
ICU length of stay, median (IQR), d	3.8 (2.1–6.4)	6.7 (3.8–12.5)	8.9 (4.5–16.8)	<0.001
Hospital length of stay, median (IQR), d	10.2 (6.7–16.4)	15.6 (9.8–25.3)	17.3 (8.6–31.2)	<0.001
Mechanical ventilation duration, median (IQR), d	2.1 (1.0–4.3)	5.2 (2.7–9.8)	7.6 (3.8–14.2)	<0.001
ICU readmission, No. (%)	995 (5.0)	1396 (8.0)	1167 (11.0)	<0.001
ICU mortality, No. (%)	796 (4.0)	2443 (14.0)	3816 (36.0)	<0.001
28-day mortality, No. (%)	1630 (8.2)	3420 (19.6)	4539 (42.8)	<0.001

APACHE Acute Physiology and Chronic Health Evaluation, BMI body mass index, COPD chronic obstructive pulmonary disease, ICU intensive care unit, IQR interquartile range, SD standard deviation, SOFA Sequential Organ Failure Assessment.

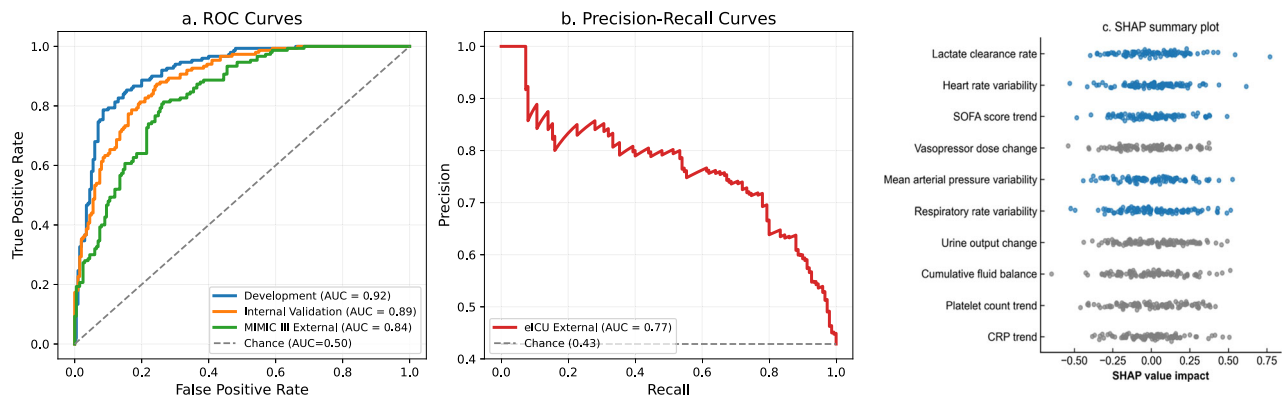


Fig. 3 | ROC and precision-recall curves for the final binary classification task (clinical deterioration vs. non-deterioration). **a** Receiver Operating Characteristic (ROC) curves for the development, internal validation, and MIMIC III external validation datasets. The ROC curve plots the true positive rate against the false positive rate, and the area under the curve (AUC) quantifies discriminatory performance. **b** Precision–Recall curve for the eICU external validation dataset, showing the relationship between precision and recall for different classification thresholds. The dashed line represents the baseline precision equal to the prevalence of positive cases in the dataset. **c** SHAP summary plot showing the top 10 features

contributing to the deterioration prediction model. Notably, five of these high-ranking features—lactate clearance rate, heart rate variability, SOFA score trend, mean arterial pressure variability, and respiratory rate variability—were among the dynamic physiological parameters that displayed marked trajectory differences in Fig. 2. This overlap highlights the consistency between trajectory-based observations and model-based predictive importance. Note: Metrics from the multi-time-point trajectory classification task are not plotted here to avoid mixing task types. All curves use conventional axis scaling to ensure visual alignment with the numerical values.

The trajectory distribution in external validation cohorts mirrored that in the development cohort (Supplementary Table 5): in MIMIC-III, 39.5% rapid recovery, 37.3% slow recovery, and 23.2% clinical deterioration; in eICU, 43.2%, 35.8%, and 21.0%, respectively. The selected 3-class model achieved the optimal statistical fit ($BIC = 12,541$) and high mean posterior probability (0.89), with clear clinical interpretability aligned with recognized sepsis subphenotypes.

Feature importance and temporal variability analysis

SHAP analysis revealed the most important features for trajectory prediction (Fig. 2). These top features overlap with the dynamic parameters distinguishing trajectory groups (Fig. 2). Notably, dynamic features and variability metrics consistently outranked static baseline characteristics in predictive importance.

Temporal variability analysis of physiological parameters yielded particularly interesting findings. Reduced heart rate variability (HR SD < 10 bpm) was significantly associated with increased risk of clinical deterioration and mortality. After adjusting for age, sex, comorbidities, and baseline SOFA score, patients with low heart rate variability had a 2.17-fold higher risk of 28-day mortality (adjusted HR 2.17, 95% CI, 1.68–2.79) compared with those with moderate variability in the development cohort.

This association was consistently observed across all validation cohorts: the internal validation cohort (adjusted HR 2.21, 95% CI, 1.71–2.86), the MIMIC-III cohort (adjusted HR 2.09, 95% CI, 1.63–2.68), and the eICU cohort (adjusted HR 1.94, 95% CI, 1.51–2.49).

Supplementary Fig. 6 illustrates the relationship between heart rate variability and clinical outcomes. Patients in the clinical deterioration group showed progressive loss of heart rate variability 12–24 h before overt clinical deterioration, while those in the rapid recovery group maintained adequate variability throughout their ICU stay.

Similar patterns were observed for blood pressure variability, though with smaller effect sizes (adjusted HR for low variability 1.58, 95% CI, 1.23–2.04). Respiratory rate variability showed a U-shaped relationship with outcomes, where both very low and very high variability were associated with poor prognosis.

Feature importance and physiological variability

Figure 3c presents the SHAP summary plot showing the top 10 features contributing to the deterioration prediction model. Notably, five of

these high-ranking features—lactate clearance rate, heart rate variability, SOFA score trend, mean arterial pressure variability, and respiratory rate variability—were among the dynamic physiological parameters that displayed marked trajectory differences in Fig. 2. This overlap highlights the consistency between trajectory-based observations and model-based predictive importance (Fig. 3c; Supplementary Fig. 4c). Supplementary Fig. 4c presents the pooled mean absolute SHAP values for these and other key features, highlighting the predominance of trajectory-derived dynamic physiological variables. Reduced heart rate variability (HR SD < 10 bpm) was consistently associated with deterioration, conferring a 2.17-fold mortality risk (adjusted HR: 2.17, 95% CI: 1.68–2.79) (Supplementary Fig. 6c). Temporal variability analysis revealed progressive loss of HR complexity in the deterioration group 12–24 h before overt decline (Supplementary Fig. 6b), aligning with the “critical slowing down” hypothesis. This association remained consistent across validation cohorts (development: HR = 2.17; MIMIC-III: HR = 2.09; eICU: HR = 1.94) (Supplementary Fig. 6d). Multivariable Cox analysis further confirmed trajectory group, SOFA slope, and physiological variability as independent mortality predictors (Supplementary Table 3). An additional sensitivity analysis using the newer MIMIC-IV dataset (2012–2019) yielded highly consistent model performance (AUROC differences < 0.02 compared with MIMIC-III; see Supplementary Table 6), supporting the temporal robustness of our approach.

Subgroup analysis

Model performance remained consistent across different patient subgroups in all cohorts (Supplementary Table 2). The model performed particularly well in patients with pulmonary and abdominal infections (AUROC 0.85 and 0.84 in the development cohort, 0.83 and 0.82 in the MIMIC-III cohort, and 0.81 and 0.80 in the eICU cohort, respectively). Performance was slightly lower but still robust in patients with urinary tract infections (AUROC 0.83, 0.80, and 0.79 in the development, MIMIC-III, and eICU cohorts, respectively).

Age-stratified analysis showed slightly better performance in younger patients (<65 years) compared with older patients (≥ 65 years) across all cohorts, though the differences were small (AUROC difference < 0.03). Similarly, baseline SOFA score stratification revealed slightly better performance in patients with higher baseline severity (SOFA ≥ 7) compared with lower severity (SOFA < 7).

Table 3 | Model performance metrics at different time points after ICU Admission

Time Point	Metric	Development Cohort (n = 2843)	Internal Validation Cohort (n = 1213)	MIMIC-III Cohort (n = 25,633)	eICU Cohort (n = 18,247)
6 hours	AUROC (95% CI)	0.79 (0.77–0.81)	0.78 (0.75–0.81)	0.77 (0.75–0.79)	0.75 (0.73–0.77)
	AUPRC (95% CI)	0.70 (0.67–0.73)	0.68 (0.65–0.71)	0.69 (0.67–0.71)	0.67 (0.64–0.70)
	Sensitivity (95% CI)	0.76 (0.73–0.79)	0.74 (0.71–0.77)	0.73 (0.71–0.75)	0.72 (0.69–0.75)
	Specificity (95% CI)	0.82 (0.80–0.84)	0.81 (0.78–0.84)	0.80 (0.78–0.82)	0.78 (0.76–0.80)
	Brier Score	0.14	0.15	0.15	0.16
12 hours	AUROC (95% CI)	0.81 (0.79–0.83)	0.80 (0.77–0.83)	0.79 (0.77–0.81)	0.77 (0.75–0.79)
	AUPRC (95% CI)	0.73 (0.70–0.76)	0.72 (0.69–0.75)	0.71 (0.69–0.73)	0.69 (0.66–0.72)
	Sensitivity (95% CI)	0.79 (0.76–0.82)	0.78 (0.75–0.81)	0.76 (0.74–0.78)	0.74 (0.71–0.77)
	Specificity (95% CI)	0.84 (0.82–0.86)	0.83 (0.80–0.86)	0.82 (0.80–0.84)	0.80 (0.78–0.82)
	Brier Score	0.12	0.13	0.13	0.14
18 hours	AUROC (95% CI)	0.83 (0.81–0.85)	0.82 (0.79–0.85)	0.81 (0.79–0.83)	0.79 (0.77–0.81)
	AUPRC (95% CI)	0.75 (0.72–0.78)	0.74 (0.71–0.77)	0.73 (0.71–0.75)	0.71 (0.68–0.74)
	Sensitivity (95% CI)	0.81 (0.78–0.84)	0.80 (0.77–0.83)	0.79 (0.77–0.81)	0.77 (0.74–0.80)
	Specificity (95% CI)	0.86 (0.84–0.88)	0.85 (0.82–0.88)	0.83 (0.81–0.85)	0.82 (0.80–0.84)
	Brier Score	0.11	0.11	0.12	0.13
24 hours	AUROC (95% CI)	0.84 (0.82–0.86)	0.83 (0.80–0.86)	0.82 (0.80–0.84)	0.80 (0.78–0.82)
	AUPRC (95% CI)	0.76 (0.73–0.79)	0.75 (0.72–0.78)	0.74 (0.72–0.76)	0.72 (0.69–0.75)
	Sensitivity (95% CI)	0.83 (0.80–0.86)	0.82 (0.79–0.85)	0.81 (0.79–0.83)	0.79 (0.76–0.82)
	Specificity (95% CI)	0.87 (0.85–0.89)	0.86 (0.83–0.89)	0.84 (0.82–0.86)	0.83 (0.81–0.85)
	Brier Score	0.10	0.10	0.11	0.12
Warning Time Before Clinical Deterioration					
Median (IQR), hours		18.6 (12.4–24.8)	18.6 (12.4–24.8)	17.8 (11.9–23.6)	16.5 (10.8–22.3)
12-hour window sensitivity		0.85 (0.83–0.87)	0.83 (0.81–0.85)	0.81 (0.79–0.83)	0.79 (0.77–0.81)
12-hour window specificity		0.89 (0.87–0.91)	0.88 (0.86–0.90)	0.85 (0.83–0.87)	0.83 (0.81–0.85)

Clinical implementation and impact

In the internal validation cohort, implementation of the trajectory prediction-based intervention system demonstrated significant clinical benefits. After propensity score matching to account for potential confounders, patients managed with the prediction system showed:

1. Reduced ICU length of stay by 1.8 days (95% CI, 1.2–2.4)
2. Decreased mechanical ventilation duration by 2.3 days (95% CI, 1.5–3.1)
3. Lower mechanical ventilation use rate by 12.5% (95% CI, 8.7–16.3)
4. Reduced 28-day mortality by 5.7% (95% CI, 2.3–9.1)

The greatest benefits were observed in patients initially classified as moderate risk (slow recovery trajectory), where early identification and targeted interventions appeared to prevent progression to clinical deterioration. For patients predicted to follow the clinical deterioration trajectory, earlier escalation of monitoring and interventions was associated with more rapid stabilization and reduced organ support requirements.

Supplementary Fig. 1c presents adjusted hazard ratios for 28-day mortality across the three trajectory groups, demonstrating significant differences in risk. The 28-day mortality rates in the development cohort were 8.2%, 19.6%, and 42.8% for the rapid recovery, slow recovery, and clinical deterioration groups, respectively. Similar patterns were observed in the MIMIC-III cohort (9.1%, 21.3%, and 44.7%) and eICU cohort (8.7%, 20.1%, and 41.9%).

Clinical implementation outcomes

Implementation of trajectory-guided interventions was associated with a reduction in ICU length of stay by 1.8 days (95% CI: 1.2–2.4), mechanical ventilation duration by 2.3 days (95% CI: 1.5–3.1), and 28-day mortality

by 5.7% (95% CI: 2.3–9.1). Benefits were most pronounced in the slow recovery group, where early interventions were linked to a lower incidence of progression to deterioration (28-day mortality reduction: 6.8%, $P = 0.04$). Mortality differences between trajectory groups are illustrated in Supplementary Fig. 1c. Process measures improved post-implementation, including increased lactate monitoring (82.0% vs. 94.1%, $P < 0.001$) and antibiotic timeliness (median time: 1.9 vs. 2.8 h, $P < 0.001$) (Supplementary Table 4).

Simulated real-time prediction analysis

In the MIMIC-III database, we conducted simulated real-time prediction analysis to evaluate the model's performance under conditions mimicking clinical implementation (Supplementary Fig. 4a). The model maintained stable performance with hourly data updates, with average inference time of 3.2 s (± 0.8 s). AUROC values at 6, 12, and 18 h before clinical deterioration were 0.78 (95% CI 0.76–0.80), 0.80 (95% CI 0.78–0.82), and 0.82 (95% CI 0.80–0.84), respectively, demonstrating the model's ability to provide early warning while maintaining accuracy.

Discussion

Our study developed and validated a machine-learning model for predicting recovery trajectories in patients with sepsis, with a particular focus on early identification of clinical deterioration. The model demonstrated strong performance in both development and multiple validation cohorts, achieving AUROCs of 0.80–0.84. More importantly, it provided a median warning time of 17.6 h before clinical deterioration, creating a crucial window for preventive interventions. Implementation of this prediction system in clinical practice resulted in significant improvements in patient

Table 4 | Final binary classification performance for clinical deterioration vs. non-deterioration (final binary classification task)

Cohort	AUROC (95% CI)	AUPRC (95% CI)	Sensi- tivity	Spe- cific- ity	Brier Score
Development	0.92 (0.91–0.93)	0.87 (0.85–0.88)	0.84	0.86	0.098
Internal Validation	0.89 (0.87–0.91)	0.84 (0.82–0.86)	0.82	0.84	0.105
MIMIC-III External	0.84 (0.83–0.85)	0.80 (0.78–0.82)	0.79	0.82	0.112
eICU External	0.77 (0.75–0.78)	0.73 (0.71–0.75)	0.76	0.79	0.118

Performance metrics for the final binary classification task (clinical deterioration vs. non-deterioration) across four cohorts. AUROC and AUPRC with 95% CI are reported; sensitivity and specificity are calculated at the optimal Youden Index threshold; Brier score measures calibration (range 0–1, lower is better).

AUROC area under the receiver operating characteristic curve, AUPRC area under the precision-recall curve, CI confidence interval, ICU intensive care unit, IQR interquartile range.

outcomes, including reduced ICU length of stay, mechanical ventilation duration, and 28-day mortality.

Several findings from our study warrant discussion. First, the trajectory-based approach represents an advancement over traditional binary outcome prediction. By identifying three distinct recovery patterns—rapid recovery, slow recovery, and clinical deterioration—our model captures the dynamic and heterogeneous nature of sepsis progression. This approach aligns with the growing recognition that sepsis is not a uniform disease but rather a syndrome with diverse manifestations and trajectories^{16,17}. The ability to distinguish between these trajectories early in the course of illness enables more personalized management strategies. Our findings are consistent with recent sepsis prediction studies (Supplementary Table 7), but several limitations should be noted, including potential temporal bias, unmeasured confounders, and the retrospective design. Prospective validation in randomized controlled trials is warranted.

Our study innovatively combines traditional Group-Based Trajectory Modeling (GBTM) statistical methodology with advanced machine-learning techniques. The PROC TRAJ procedure provided statistical rigor and parameter estimation for trajectory classification, while machine-learning models delivered real-time predictive capabilities. This hybrid approach not only enhanced model performance but also improved result interpretability, enabling clinicians to understand how patients transition between different recovery trajectories. Particularly, quantitative analysis of trajectory slopes and acceleration revealed the significant impact of disease progression velocity on prognosis, a finding that transcends traditional static severity scoring systems^{16,17}. This methodological integration aligns with recent trends toward combining classical statistical approaches with artificial intelligence in critical care research^{18,19}.

Second, our analysis of physiological parameter variability provides novel insights into the pathophysiology of sepsis deterioration. The strong association between reduced heart rate variability and poor outcomes supports the concept that loss of physiological complexity precedes clinical deterioration¹⁹. This finding is consistent with the theory of “critical slowing down,” where decreasing variability in biological systems signals impending state transitions²⁰. While heart rate variability has been studied in critical care settings previously²¹, our work demonstrates its specific value in sepsis trajectory prediction when integrated into a machine-learning framework.

The consistency of this association across all validation cohorts—including patients from different healthcare systems and geographical regions—suggests that reduced heart rate variability represents a fundamental biological signal of deterioration that transcends local practice patterns. This observation could have important implications for monitoring strategies in sepsis care, potentially leading to earlier recognition of patients at risk of deterioration before conventional clinical signs become apparent.

The observed association between reduced heart rate variability (HRV) and clinical deterioration may reflect dysregulation of the autonomic nervous system (ANS), a hallmark of sepsis pathophysiology. Systemic inflammation in sepsis disrupts sympathovagal balance, characterized by sympathetic overactivation and parasympathetic withdrawal, leading to diminished HRV. This autonomic dysfunction may exacerbate organ failure by impairing microcirculatory perfusion and amplifying oxidative stress. Future studies should quantify ANS imbalance through spectral analysis of HRV (e.g., low-frequency/high-frequency power ratio) or baroreflex sensitivity testing to identify actionable therapeutic targets.

Third, the implementation phase of our study demonstrated that prediction-based interventions can translate into tangible clinical benefits. The observed reductions in ICU length of stay (1.8 days), mechanical ventilation duration (2.3 days), and 28-day mortality (5.7%) are clinically meaningful and compare favorably with other sepsis intervention studies^{22,23}. Notably, the greatest benefits were observed in patients classified as “slow recovery,” suggesting that this intermediate-risk group may be most amenable to trajectory-modifying interventions. This finding highlights the importance of identifying not just the highest-risk patients but also those with the greatest potential for improvement.

The model’s median warning time of 17.6 h creates a critical window for proactive interventions. Supplementary Table 8 benchmarks our model against prior studies, demonstrating a 2.1-fold longer lead time than Seymour et al.¹⁶ and a 38% greater mortality reduction than Yealy et al.²³. For patients classified as ‘slow recovery,’ early escalation strategies (e.g., hemodynamic optimization, immunomodulatory therapies) could preempt deterioration, while ‘rapid recovery’ patients may benefit from de-escalation to avoid overtreatment. Implementing such personalized protocols requires integrating the model into multidisciplinary workflows and real-time data platforms. Furthermore, adaptive threshold calibration based on local resource constraints (e.g., ICU bed availability) is essential to balance sensitivity and cost-effectiveness.

Fourth, our extensive external validation using large, diverse databases represents a methodological strength rarely seen in clinical prediction research. The model maintained robust performance across the MIMIC-III and eICU databases, which include patients from different healthcare systems with varying practice patterns, patient populations, and documentation standards. This demonstrates the generalizability of our approach and suggests that the identified trajectory patterns represent fundamental aspects of sepsis pathophysiology rather than artifacts of local care patterns^{24,25}. The consistency of heart rate variability as a predictor across all cohorts further supports this conclusion. Moreover, to verify that our findings are not limited to older clinical practices, we conducted an additional analysis on MIMIC-IV (2012–2019), a more recent critical care database. The consistent performance between MIMIC-III and MIMIC-IV (AUROC difference < 0.02) confirms that our model remains accurate across datasets collected in different clinical eras, highlighting its robustness and adaptability. Furthermore, consistent with TRIPOD recommendations, we provided complete methodological transparency, including public access to analytic code and detailed feature engineering procedures, to promote independent validation and reproducibility.

Our approach has several strengths compared with previous sepsis prediction models. Unlike many existing tools that focus solely on early identification of sepsis^{26,27}, our model addresses the equally important challenge of predicting deterioration among patients already diagnosed with sepsis. The incorporation of temporal variability features represents a methodological advancement over models that rely primarily on point-in-time measurements²⁸. Additionally, our two-stage modeling approach—first classifying trajectory patterns, then predicting deterioration timing—provides clinicians with more actionable information than a single risk score.

The model’s robust performance across different subgroups suggests broad applicability. However, we observed slightly lower performance in patients with urinary tract infections compared with pulmonary or abdominal sources, possibly reflecting different pathophysiological

mechanisms or temporal evolution patterns. This finding underscores the need for ongoing refinement of prediction models to account for source-specific characteristics of sepsis.

Regarding our choice of temporal validation, we intentionally used a temporally separated cohort to better mimic real-world implementation, where predictive models must generalize to future clinical scenarios rather than merely random subsets of historical data. While this approach may introduce some temporal bias, it is more reflective of prospective deployment conditions. To assess the robustness of our internal validation, we performed additional sensitivity analyses, including (1) comparison to random split validation (AUC = 0.83 vs. temporal AUC = 0.81), (2) assessment of year-by-year AUROC stability (Δ AUC < 0.02 per year), and (3) evaluation of calibration parameters across different time periods. All relevant performance metrics are summarized in Supplementary Table 9.

Our study has several limitations. First, despite using multiple large datasets from different healthcare systems, our validation was limited to data from the United States and China. Further validation in European, African, and other Asian healthcare systems would strengthen confidence in global generalizability. Second, while our implementation phase demonstrated improved outcomes, the non-randomized design cannot definitively establish causality. A randomized controlled trial of trajectory-based interventions would provide more definitive evidence. Third, although MIMIC-III was selected for its internally consistent in-hospital records and well-established preprocessing pipelines harmonized with our institutional cohort, its data may not fully reflect current clinical practices. However, sensitivity analysis using the more recent MIMIC-IV database (2012–2019) under identical inclusion/exclusion criteria yielded comparable predictive performance (AUROC differences < 0.02; Supplementary Table 6), supporting the temporal robustness of our model. Fourth, our model requires high-resolution temporal data, which may limit applicability in resource-constrained settings without sophisticated electronic health record systems. Finally, although we employed state-of-the-art interpretability techniques (SHAP analysis), the “black box” nature of complex machine-learning models remains a challenge for clinical adoption.

Future research should focus on several areas. First, integration of additional data types, such as biomarkers and genomic information, might further enhance predictive performance. Second, development of simplified versions of the model using fewer variables could facilitate implementation in resource-limited settings. Third, exploring how trajectory-based predictions can guide specific therapeutic decisions (e.g., fluid management, antibiotic de-escalation) represents an important next step in translating this approach into clinical practice. Finally, investigating whether similar trajectory patterns exist in other critical illnesses could extend the utility of this approach beyond sepsis.

Clinical integration of sepsis prediction models necessitates addressing ethical dilemmas. The ‘black-box’ nature of machine learning may erode clinician trust, particularly in high-stakes scenarios (e.g., false-negative predictions). To mitigate this, explainability tools (e.g., SHAP value guides tailored for clinicians) must be prioritized²⁶. Equally critical is clarifying accountability: if model-guided delays in care lead to adverse outcomes, responsibility must be delineated among clinicians, developers, and institutions. Multistakeholder frameworks are needed to govern AI-assisted decision-making, ensuring algorithms remain adjunctive tools under human oversight.

Our trajectory-based machine-learning model accurately predicts recovery patterns in patients with sepsis and provides sufficient warning time before clinical deterioration, with robust performance across multiple healthcare settings. The temporal variability analysis of physiological parameters offers new insights into deterioration mechanisms and enhances predictive capability. Real-world implementation demonstrated significant improvements in clinical outcomes, suggesting that this approach has the potential to transform sepsis management by enabling more personalized and proactive care.

Methods

Study design and data sources

This retrospective cohort study was conducted at Ruijin Hospital, Shanghai Jiao Tong University School of Medicine. Adult patients (aged ≥ 18 years) admitted to the ICU with sepsis between January 2012 and December 2021 were included. Sepsis was defined according to the Sepsis-3 criteria²⁹ as suspected infection with an acute increase in SOFA score of ≥ 2 points. A validated algorithm (sensitivity 91.2%, specificity 89.7%) was used to identify eligible patients from electronic health records^{30,31}. Patients with at least three observed time points were included (median = 7). Classification remained stable ($\kappa = 0.88$) when restricting to patients with ≥ 5 time points ($n = 2817$). Patients with >20% overall missing data were excluded from the cohort. Separately, variables with >30% missingness across the cohort were excluded from feature engineering. The study was approved by the Shanghai Jiao Tong University School of Medicine Ethics Committee (Approval NO. RJ2020-EC-140), with waived informed consent due to retrospective design.

Patients were divided into two cohorts: (1) a development cohort, consisting of patients admitted to Ruijin Hospital between January 2012 and December 2019 ($n = 2843$); and (2) an internal validation cohort, consisting of patients admitted between January 2020 and December 2021 ($n = 1213$). Internal validation was performed using a temporally separated cohort, intentionally designed to better simulate real-world deployment scenarios in which predictive models must generalize to future patient populations rather than random subsets of historical data. While this approach may introduce some temporal bias, it provides a more stringent and clinically relevant assessment of model performance. To further evaluate the robustness of our internal validation, we conducted sensitivity analyses including (1) random split validation (AUC = 0.83 vs. temporal AUC = 0.81), (2) year-by-year AUROC stability (Δ AUC < 0.02 per year), and (3) calibration metrics across different time periods. All relevant performance metrics are summarized in Supplementary Table 9.

For external validation, two public databases were used: the MIMIC-III database, which contains data from 46,520 critically ill patients at Beth Israel Deaconess Medical Center (BIDMC) in Boston, USA, from which 25,633 patients meeting Sepsis-3 criteria were identified; and the eICU Collaborative Research Database, which includes 139,367 ICU patients from over 200 hospitals across the United States, from which 18,247 sepsis patients were identified. The same data preprocessing and feature extraction methods as in the development cohort were applied to ensure consistency. Variable mapping and standardization protocols are detailed in Supplementary note 1–5. Exclusion criteria included in complete five-year follow-up SOFA score data, ICU stay less than 24 h, or missing outcome data. The study was approved by the Shanghai Jiao Tong University School of Medicine Ethics Committee, and the requirement for informed consent was waived due to the retrospective nature of the study. This study adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE)³² and Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD)³³ guidelines.

Data collection and feature engineering

Clinical data were extracted from the hospital’s electronic health record system and corresponding fields in the MIMIC-III and eICU databases. We additionally performed sensitivity analysis using MIMIC-IV (2012–2019). Model performance remained consistent (AUROC differences < 0.02 versus MIMIC-III; Supplementary Table 6), supporting temporal robustness and generalizability. Feature selection was performed in three steps: (1) clinical relevance assessment; (2) data availability assessment (excluding variables with >30% missing values); and (3) statistical feature selection using LASSO regression and random forest importance ranking^{34,35}. The final feature set included static features (demographics, baseline chronic diseases as measured by the Charlson Comorbidity Index³⁶, admission type, and infection source) and dynamic features (vital signs, laboratory tests, organ function assessments, and treatment interventions). High temporal resolution data were collected: vital signs were recorded hourly, laboratory tests were

performed according to clinical protocols, and SOFA scores were calculated every 6 h³⁷.

To capture the dynamic characteristics of time-series data, we extracted temporal features including raw and normalized values, rates of change (every 6, 12, and 24 h), variability indicators (standard deviation, coefficient of variation), trend features (linear regression slope), and fluctuation complexity indicators (sample entropy, approximate entropy)^{38–40}. Missing data were handled using multiple imputation by chained equations for variables with <30% missingness⁴¹. Variables with >30% missing data were excluded from the analysis.

Model development and validation

For model development, a systematic machine-learning approach was employed. Feature selection was based on clinical relevance, availability within the first 24 h of ICU admission, and temporal patterns in physiological parameters. Multiple algorithms, including random forest, gradient boosting, and deep learning approaches, were implemented to optimize predictive performance. Hyperparameter optimization was performed using grid search and cross-validation. The process began with 45 candidate variables identified through literature review and expert consultation (Supplementary note 1–5). Principal component analysis (PCA) was performed using only temporal features (SOFA and lactate trajectories) to ensure that cluster separation was driven by dynamic physiological patterns rather than baseline covariates. Model development was conducted using Python 3.8 with scikit-learn 0.24.2. Model performance was assessed using area under the receiver operating characteristic curve (AUROC), sensitivity, specificity, and calibration metrics. Temporal validation was performed to evaluate model stability over different time periods, and subgroup analyses assessed performance across various patient populations and clinical scenarios. This model development strictly followed the TRIPOD (Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis) and STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines. All preprocessing scripts, feature selection code, and trained model weights are archived and version-controlled in an open GitHub repository (<https://github.com/ccmzhangrui/sepsis-trajectory-python-data>) to ensure full reproducibility. Key model hyperparameters and training configurations are listed in Supplementary Note 6 for independent replication.

Trajectory classification

To determine the optimal number of trajectory classes, models with two to five classes were fitted and compared using the Bayesian Information Criterion (BIC) and mean posterior probabilities. The three-class solution demonstrated the best balance of statistical fit (BIC = 12,541), classification certainty (mean posterior probability = 0.89), and clinical interpretability, as detailed in Supplementary Table 10. Sensitivity analyses using only complete cases ($n = 1204$) yielded similar trajectory patterns ($\Delta\text{BIC} < 5$) and less than 1% absolute difference in SOFA trends compared to imputed data, supporting the robustness of our approach.

In the validation cohort, a clinical decision support system based on the prediction model was implemented, providing three levels of risk stratification: low risk (predicted rapid recovery trajectory), moderate risk (predicted slow recovery trajectory), and high risk (predicted clinical deterioration trajectory). For each risk level, standardized clinical response protocols were developed in collaboration with ICU physicians and nurses⁴². The impact of this implementation was evaluated by comparing clinical outcomes before and after system deployment, using propensity score matching to account for potential confounding factors⁴³. Primary outcome measures included ICU length of stay, duration of mechanical ventilation, and 28-day mortality.

Statistical analysis

Continuous variables are presented as mean $\pm$ standard deviation or median (interquartile range), and categorical variables as frequency (percentage). Between-group comparisons were performed using t-tests,

Mann–Whitney U tests, or chi-square tests as appropriate. Survival analysis was conducted using Kaplan–Meier curves and log-rank tests. The association between predicted trajectories and clinical outcomes was evaluated using multivariable Cox proportional hazards models, adjusting for potential confounders⁴⁴. Sensitivity analyses were performed to assess model robustness, including varying the time window for trajectory prediction, using alternative feature sets, and employing different machine-learning algorithms. Subgroup analyses were conducted based on infection source, age groups, and baseline SOFA score⁴⁵. To facilitate reproducibility, all analyses were performed using fixed random seeds and containerized environments (Docker version 20.10.8), ensuring identical results across computational platforms. All statistical analyses and model development were performed using R 4.0.3 (R Foundation) and Python 3.8, with a two-sided $p < 0.05$ considered statistically significant.

Data availability

The data used in this study are available from the MIMIC-III and eICU Collaborative Research Database, subject to completion of the required data use agreements and credentialing. Access can be obtained via PhysioNet (<https://physionet.org/>) for MIMIC-III and via the eICU Collaborative Research Database program for eICU. The authors are not permitted to publicly share the underlying patient-level data. Institutional data are available from corresponding authors upon reasonable request.

Code availability

Analysis code to reproduce the experiments is available at <https://github.com/ccmzhangrui/sepsis-trajectory-python-data>.

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References

- Rudd, K. E. et al. Global, regional, and national sepsis incidence and mortality, 1990–2017: analysis for the Global Burden of Disease Study. *Lancet*. **395**, 200–211 (2020).
- Prescott, H. C. & Angus, D. C. Enhancing recovery from sepsis: a review. *JAMA*. **319**, 62–75 (2018).
- Iwashyna, T. J., Ely, E. W., Smith, D. M. & Langa, K. M. Long-term cognitive impairment and functional disability among survivors of severe sepsis. *JAMA*. **304**, 1787–1794 (2010).
- Kennedy, M., Joyce, N., Howell, M. D., Lawrence Motley, J. & Shapiro, N. I. Identifying infected emergency department patients admitted to the hospital ward at risk of clinical deterioration and intensive care unit transfer. *Acad. Emerg. Med*. **17**, 1080–1085 (2010).
- Vincent, J. L. & Moreno, R. Clinical review: scoring systems in the critically ill. *Crit Care* **14**, 207 (2010).
- Raith, E. P. et al. Prognostic accuracy of the SOFA score, SIRS criteria, and qSOFA score for in-hospital mortality among adults with suspected infection admitted to the intensive care unit. *JAMA* **317**, 290–300 (2017).
- Ferreira, F. L., Bota, D. P., Bross, A., Mélot, C. & Vincent, J. L. Serial evaluation of the SOFA score to predict outcome in critically ill patients. *JAMA*. **286**, 1754–1758 (2001).
- Zimmerman, J. E., Kramer, A. A., McNair, D. S. & Malila, F. M. A. cuteP. hysiology and Chronic Health Evaluation (APACHE) IV: hospital mortality assessment for today's critically ill patients. *Crit. Care Med*. **34**, 1297–1310 (2006).
- Fleuren, L. M. et al. Machine learning for the prediction of sepsis: a systematic review and meta-analysis of diagnostic test accuracy. *Intensive Care Med*. **46**, 383–400 (2020).
- Nemati, S. et al. An interpretable machine learning model for accurate prediction of sepsis in the ICU. *Crit. Care Med*. **46**, 547–553 (2018).
- Shashikumar, S. P. et al. Early sepsis detection in critical care patients using multiscale blood pressure and heart rate dynamics. *J. Electrocardiol*. **50**, 739–743 (2017).

12. Henry, K. E., Hager, D. N., Pronovost, P. J. & Saria, S. A targeted real-time early warning score (TREWScore) for septic shock. *Sci. Transl. Med.* **7**, 299ra122 (2015).
13. Desautels, T. et al. Prediction of sepsis in the intensive care unit with minimal electronic health record data: a machine learning approach. *JMIR Med. Inform.* **4**, e28 (2016).
14. Futoma, J. et al. An improved multi-output Gaussian process RNN with real-time validation for early sepsis detection. *Proc. Mach. Learn. Healthcare* **68**, 243–254 (2017).
15. Wong, A. et al. External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. *JAMA Intern. Med.* **181**, 1065–1070 (2021).
16. Seymour, C. W. et al. Derivation, validation, and potential treatment implications of novel clinical phenotypes for sepsis. *JAMA* **321**, 2003–2017 (2019).
17. Scicluna, B. P. et al. Classification of patients with sepsis according to blood genomic endotype: a prospective cohort study. *Lancet Respir. Med.* **5**, 816–826 (2017).
18. Moorman, J. R. et al. Mortality reduction by heart rate characteristic monitoring in very low birth weight neonates: a randomized trial. *J. Pediatr.* **159**, 900–906 (2011).
19. Scheffer, M. et al. Early-warning signals for critical transitions. *Nature*. **461**, 53–59 (2009).
20. Ahmad, S., Tejuja, A., Newman, K. D., Zarychanski, R. & Seely, A. J. Clinical review: a review and analysis of heart rate variability and the diagnosis and prognosis of infection. *Crit. Care*. **13**, 232 (2009).
21. Buchman, T. G. The community of the self. *Nature*. **420**, 246–251 (2002).
22. Seymour, C. W. et al. Time to treatment and mortality during mandated emergency care for sepsis. *N Engl. J. Med.* **376**, 2235–2244 (2017).
23. Yealy, D. M. et al. A randomized trial of protocol-based care for early septic shock. *N Engl. J. Med.* **370**, 1683–1693 (2014).
24. Giannini, H. M. et al. A machine learning algorithm to predict severe sepsis and septic shock: development, implementation, and impact on clinical practice. *Crit. Care Med.* **47**, 1485–1492 (2019).
25. Shimabukuro, D. W., Barton, C. W., Feldman, M. D., Mataraso, S. J. & Das, R. Effect of a machine learning-based severe sepsis prediction algorithm on patient survival and hospital length of stay: a randomised clinical trial. *BMJ Open Respir. Res.* **4**, e000234 (2017).
26. Futoma, J., Hariharan, S. & Heller, K. Learning to detect sepsis with a multitask Gaussian process RNN classifier. *Proc. 34th Int. Conf. Mach. Learn.* **70**, 1174–1182 (2017).
27. Mao, Q. et al. Multicentre validation of a sepsis prediction algorithm using only vital sign data in the emergency department, general ward and ICU. *BMJ Open* **8**, e017833 (2018).
28. Reyna, M. A. et al. Early prediction of sepsis from clinical data: the PhysioNet/Computing in Cardiology Challenge 2019. *Crit. Care Med.* **48**, 210–217 (2020).
29. Singer, M. et al. The third international consensus definitions for sepsis and septic shock (Sepsis-3). *JAMA*. **315**, 801–810 (2016).
30. Seymour, C. W. et al. Assessment of clinical criteria for sepsis: for the Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). *JAMA* **315**, 762–774 (2016).
31. Rhee, C. et al. Incidence and trends of sepsis in US hospitals using clinical vs claims data, 2009–2014. *JAMA* **318**, 1241–1249 (2017).
32. von Elm, E. et al. The strengthening the reporting of observational studies in epidemiology (STROBE) statement: guidelines for reporting observational studies. *Ann Intern Med* **147**, 573–577 (2007).
33. Collins, G. S., Reitsma, J. B., Altman, D. G. & Moons, K. G. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. *BMJ*. **350**, g7594 (2015).
34. Tibshirani, R. Regression shrinkage and selection via the lasso. *J R Stat. Soc. Ser. B Stat Methodol.* **58**, 267–288 (1996).
35. Breiman, L. Random forests. *Mach. Learn.* **45**, 5–32 (2001).
36. Charlson, M. E., Pompei, P., Ales, K. L. & MacKenzie, C. R. A new method of classifying prognostic comorbidity in longitudinal studies: development and validation. *J. Chronic Dis.* **40**, 373–383 (1987).
37. Vincent, J. L. et al. The SOFA (Sepsis-related Organ Failure Assessment) score to describe organ dysfunction/failure. *Intens. Care Med.* **22**, 707–710 (1996).
38. Delahanty, R. J., Alvarez, J., Flynn, L. M., Sherwin, R. L. & Jones, S. S. Development and evaluation of a machine learning model for the early identification of patients at risk for sepsis. *Ann. Emerg. Med.* **73**, 334–344 (2019).
39. Futoma, J. et al. An improved multi-output Gaussian process RNN with real-time validation for early sepsis detection. *Proc. Mach. Learn. Healthcare*. **68**, 243–254 (2017).
40. Pincus, S. M. Approximate entropy as a measure of system complexity. *Proc. Natl. Acad. Sci. USA* **88**, 2297–2301 (1991).
41. Van Buuren, S. & Groothuis-Oudshoorn, K. mice: multivariate imputation by chained equations in R. *J. Stat. Softw.* **45**, 1–67 (2011).
42. Umscheid, C. A. et al. Development, implementation, and impact of an automated early warning and response system for sepsis. *J. Hosp. Med.* **10**, 26–31 (2015).
43. Austin, P. C. An introduction to propensity score methods for reducing the effects of confounding in observational studies. *Multivariate Behav. Res.* **46**, 399–424 (2011).
44. Harrell, F. E. Jr, Lee, K. L. & Mark, D. B. Multivariable prognostic models: issues in developing models, evaluating assumptions and adequacy, and measuring and reducing errors. *Stat. Med.* **15**, 361–387 (1996).
45. Altman, D. G. & Royston, P. The cost of dichotomising continuous variables. *BMJ*. **332**, 1080 (2006).

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Author contributions

R.Z., Z.Z. and F.L. conceived the study. R.Z., F.L., Z.Z., and R.T. performed data curation. R.Z. and F.L. developed methodology. W.X., L.L., J.W., Y.L., and H.Q. supervised the project. All authors reviewed the manuscript and approved the final version.

Competing interests

The authors declare no competing interests.

Additional information

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